Class 11: AlphaFold

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Background

In this hands-on session we will utilize AlphaFold to predict protein structure from sequence.

Without the aid of such approaches, it can take years of expensive laboratory work to determine the structure of just one protein. With AlphaFold we can now accurately compute a typical protein structure in as little as ten minutes.

The PDB database (the main repository of experimental structures) only has ~**250 thousand** structures (we saw this in the last lab). The main protein sequence database over **200 million** sequences! Only 0.125% of known sequences have known structure - this is called the "structure knowledge gap".

```
(250000 / 200000000 ) *100
```

```
[1] 0.125
```

- Structures are much harder to determine than sequences
- They are expensive (on average ~$1 million each)
- They take on average 3-5 years to solve!

EBI AlphaFold Database

The EBI has a database of pre-computed ALphaFold (AF) models called AFDB. This is growing overtime and can be useful to check before running AF ourselves

Running AlphaFold

We can download and run locally (on our own computers) but we need a GPU. Or we can use "cloud" computing to run this on someone elses computers :-)

We will use ColabFold < https://github.com/sokrypton/ColabFold >

We previously found there was no AFDB entry for our HIV sequence:

```
>HIV-Pr-Dimer
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQ
```

Here we will use AlphaFold2_mmseqs2

Custom analysis of resulting models

We will move our AlphaFold results into RStudio

```
results_dir <- "hivpr_23119/"
```

```
pdb_files <- list.files(path=results_dir,
                        pattern="*.pdb",
                        full.names = TRUE)
basename(pdb_files)
```

```
[1] "hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

Here we will see the model sequences and all its alignments

```r
library(bio3d)

# Read all data from Models
#  and superpose/fit coords
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
hivpr_23119//hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb
.....

Extracting sequences

pdb/seq: 1   name: hivpr_23119//hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4
pdb/seq: 2   name: hivpr_23119//hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1
pdb/seq: 3   name: hivpr_23119//hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5
pdb/seq: 4   name: hivpr_23119//hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2
pdb/seq: 5   name: hivpr_23119//hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3
```

We will now calculate the structural distance betwee coordinate sets using RMSD

```r
rd <- rmsd(pdbs, fit=T)
```

```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 198 non NA positions
```

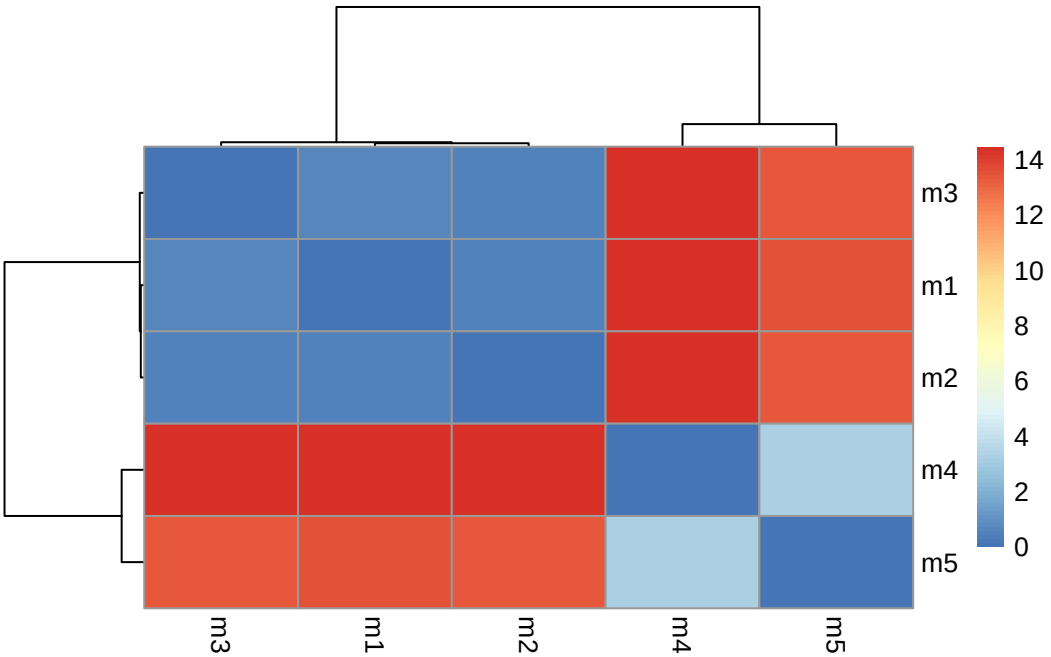
```r
range(rd)
```

```
[1]  0.000 14.457
```

Create a pheatmap of these coordinates

```r
library(pheatmap)

colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```

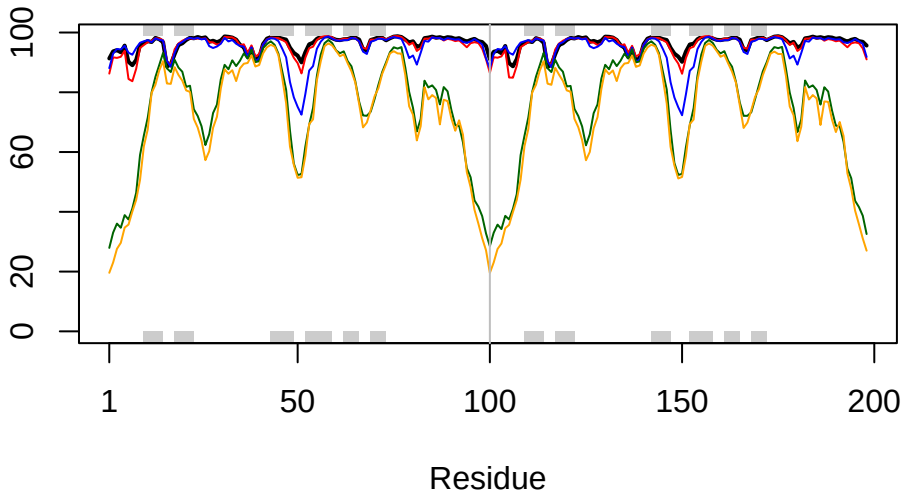


We can now plot the pLDDT values into a PDB files and add our desired colors into the graph

```
pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdb)
points(pdbs$b[2,], typ="l", col="red")
points(pdbs$b[3,], typ="l", col="blue")
points(pdbs$b[4,], typ="l", col="darkgreen")
points(pdbs$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```



Fix the superposition of the model creating a rigid core using `core.find()`

```
core <-core.find(pdbs)
```

```
core size 197 of 198  vol = 5754.474
core size 196 of 198  vol = 4572.526
core size 195 of 198  vol = 2115.143
core size 194 of 198  vol = 1525.93
core size 193 of 198  vol = 1390.829
core size 192 of 198  vol = 1280.595
core size 191 of 198  vol = 1171.652
core size 190 of 198  vol = 1082.83
core size 189 of 198  vol = 1047.706
core size 188 of 198  vol = 1021.701
core size 187 of 198  vol = 1001.511
core size 186 of 198  vol = 982.408
core size 185 of 198  vol = 965.11
core size 184 of 198  vol = 945.291
core size 183 of 198  vol = 922.93
core size 182 of 198  vol = 879.676
core size 181 of 198  vol = 856.189
core size 180 of 198  vol = 811.173
core size 179 of 198  vol = 789.408
```

```
core size 178 of 198  vol = 764.613
core size 177 of 198  vol = 743.895
core size 176 of 198  vol = 720.032
core size 175 of 198  vol = 699.404
core size 174 of 198  vol = 679.546
core size 173 of 198  vol = 662.076
core size 172 of 198  vol = 640.751
core size 171 of 198  vol = 619.091
core size 170 of 198  vol = 600.089
core size 169 of 198  vol = 582.484
core size 168 of 198  vol = 560.512
core size 167 of 198  vol = 526.05
core size 166 of 198  vol = 494.758
core size 165 of 198  vol = 477.336
core size 164 of 198  vol = 459.201
core size 163 of 198  vol = 444.277
core size 162 of 198  vol = 428.758
core size 161 of 198  vol = 415.985
core size 160 of 198  vol = 401.762
core size 159 of 198  vol = 385.62
core size 158 of 198  vol = 371.755
core size 157 of 198  vol = 357.766
core size 156 of 198  vol = 345.196
core size 155 of 198  vol = 331.481
core size 154 of 198  vol = 317.372
core size 153 of 198  vol = 303.658
core size 152 of 198  vol = 291.234
core size 151 of 198  vol = 276.942
core size 150 of 198  vol = 265.069
core size 149 of 198  vol = 254.695
core size 148 of 198  vol = 244.929
core size 147 of 198  vol = 234.256
core size 146 of 198  vol = 223.932
core size 145 of 198  vol = 214.812
core size 144 of 198  vol = 204.565
core size 143 of 198  vol = 196.615
core size 142 of 198  vol = 182.849
core size 141 of 198  vol = 173.456
core size 140 of 198  vol = 166.316
core size 139 of 198  vol = 157.971
core size 138 of 198  vol = 151.565
core size 137 of 198  vol = 145.31
core size 136 of 198  vol = 139.486
```

```
core size 135 of 198  vol = 133.291
core size 134 of 198  vol = 127.405
core size 133 of 198  vol = 121.952
core size 132 of 198  vol = 116.445
core size 131 of 198  vol = 108.879
core size 130 of 198  vol = 102.219
core size 129 of 198  vol = 94.88
core size 128 of 198  vol = 88.654
core size 127 of 198  vol = 83.357
core size 126 of 198  vol = 79.603
core size 125 of 198  vol = 77.313
core size 124 of 198  vol = 74.084
core size 123 of 198  vol = 70.385
core size 122 of 198  vol = 66.687
core size 121 of 198  vol = 62.458
core size 120 of 198  vol = 59.279
core size 119 of 198  vol = 56.355
core size 118 of 198  vol = 53.213
core size 117 of 198  vol = 49.293
core size 116 of 198  vol = 46.5
core size 115 of 198  vol = 42.868
core size 114 of 198  vol = 39.993
core size 113 of 198  vol = 36.701
core size 112 of 198  vol = 34.827
core size 111 of 198  vol = 32.668
core size 110 of 198  vol = 29.561
core size 109 of 198  vol = 26.953
core size 108 of 198  vol = 24.475
core size 107 of 198  vol = 21.998
core size 106 of 198  vol = 20.512
core size 105 of 198  vol = 18.384
core size 104 of 198  vol = 16.871
core size 103 of 198  vol = 14.911
core size 102 of 198  vol = 13.128
core size 101 of 198  vol = 11.113
core size 100 of 198  vol = 9.381
core size 99 of 198  vol = 7.991
core size 98 of 198  vol = 6.99
core size 97 of 198  vol = 6.059
core size 96 of 198  vol = 5.128
core size 95 of 198  vol = 4.231
core size 94 of 198  vol = 3.692
core size 93 of 198  vol = 3.126
```

```
 core size 92 of 198  vol = 2.754
 core size 91 of 198  vol = 2.271
 core size 90 of 198  vol = 1.642
 core size 89 of 198  vol = 1.065
 core size 88 of 198  vol = 0.644
 core size 87 of 198  vol = 0.444
 FINISHED: Min vol ( 0.5 ) reached
```

```r
core.inds <- print(core, vol=0.5)
```

```
# 88 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     8  95     88
```

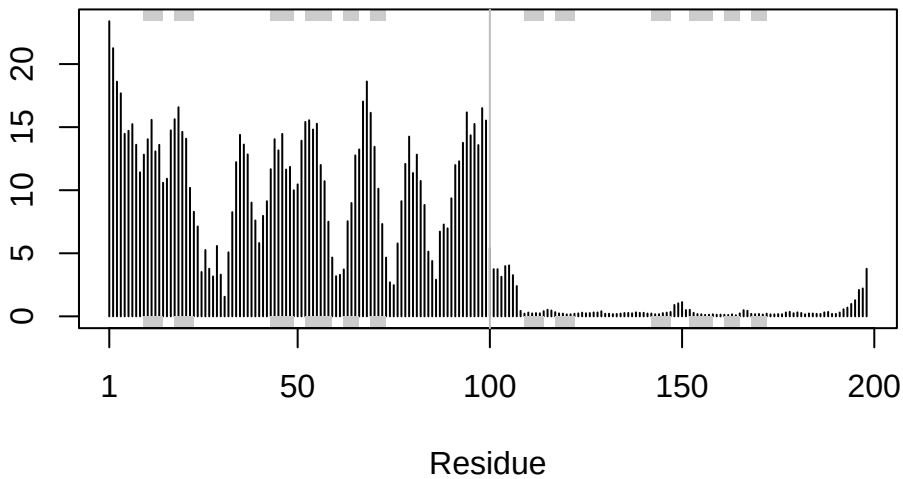
```r
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")
```

Now that we have superposed coordinates we can open our structure from Mol*

Here we will examine RMSF and its positions

```r
rf <- rmsf(xyz)

plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")
```



Predicted Alignment Error for domains

Read files and predict alignment error for domains using `jsonlite()`

```
library(jsonlite)

pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                         full.names = TRUE)
```

For example we will read models 1 and 5 here:

```
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

attributes(pae1)
```

```
$names
[1] "plddt"   "max_pae" "pae"     "ptm"     "iptm"
```

```r
head(pae1$plddt)
```

```
[1] 91.31 93.69 94.12 93.38 95.62 90.19
```

Calculate the max PAE value

```r
pae1$max_pae
```

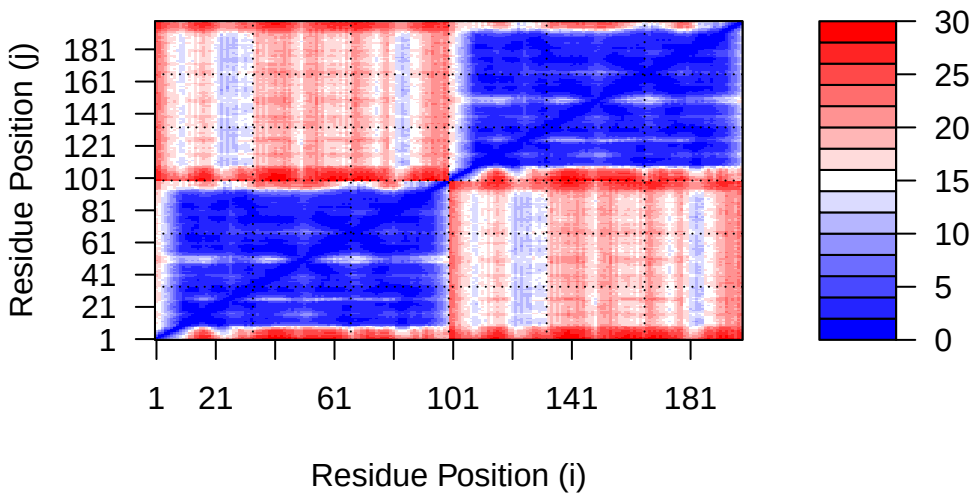
```
[1] 12.57812
```

```r
pae5$max_pae
```

```
[1] 29.82812
```

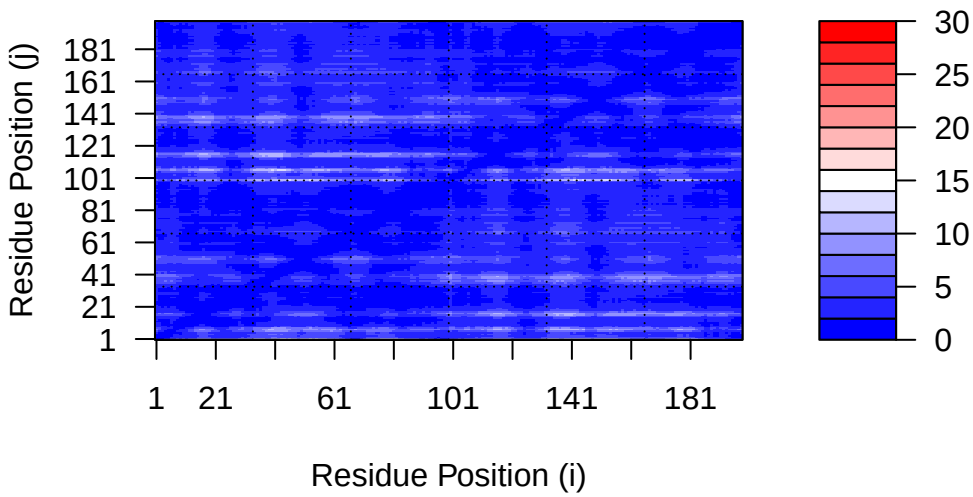
We can now plot this

```r
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



Now we will plot using the same z range

```
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



Residue conservation from alignment file

```
aln_file <- list.files(path=results_dir,
                       pattern=".a3m$",
                        full.names = TRUE)
aln_file
```

```
[1] "hivpr_23119//hivpr_23119.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

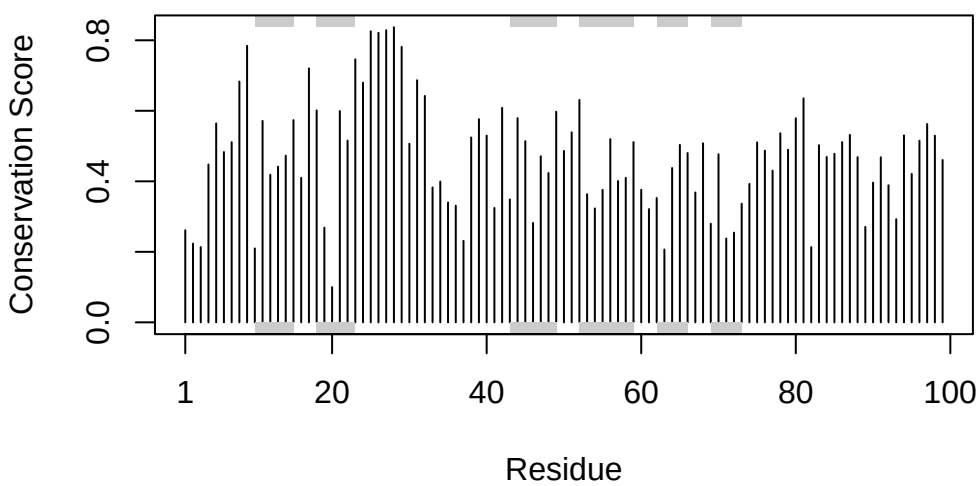
Calculate how many sequences are in the alignment

```
dim(aln$ali)
```

```
[1] 5397  132
```

Use `conserv()` to score the residue conservation

```
sim <- conserv(aln)
plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```



View parts where there is a high cutoff value in the consensus sequence

```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

```r
m1.pdb <- read.pdb(pdb_files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```